Review of "The Incompleteness of the Instantaneous State and the Status of Chapman-Kolmogorov Divisibility"

1 Overall assessment

The manuscript presents a concise and clearly organised conceptual argument concerning the status of instantaneous states and Chapman-Kolmogorov (CK) divisibility. Its principal contribution is to bring together three familiar considerations–the punctured-neighbourhood definition of a limit, Planck-scale measurement bounds, and ultraviolet behaviour in quantum field theory–and use them jointly to motivate indivisible stochastic dynamics. The individual ingredients are not new, as the manuscript acknowledges, but their proposed connection to the completeness of instantaneous states and CK divisibility constitutes a recognisable interpretative synthesis.

The manuscript is unusually clear about the intended target of its argument and usefully distinguishes the existence of a mathematical point-state from the stronger claim that such a state is dynamically complete. It also separates its three evidential strands carefully and acknowledges important limitations, including that the argument from limits does not prove incompleteness and that physics below the Planck scale remains unknown. The distinction between the evolution of a probability law and the stochastic realisation of an individual outcome is also potentially useful, provided that the scope of the cited stochastic-quantum correspondence is stated precisely.

The central conclusion, however, is not yet established. The inability to measure or physically resolve an exact temporal point does not by itself imply that the mathematical state assigned to that point is dynamically incomplete. Similarly, singularities of point-local quantities in quantum field theory do not imply that transition maps cannot factor through an intermediate time. CK divisibility is a property of a family of transition maps; it does not require an observer to measure an exact instant or the intermediate state to be constructed experimentally by shrinking an averaging interval. A theory may therefore possess exact point-indexed states and divisible propagators even when no physical clock has unlimited temporal resolution.

The arguments presently support, at most, an epistemic or operational limitation on pointwise state determination. They do not yet establish ontological incompleteness, failure of dynamical sufficiency, or generic CK indivisibility. These different meanings of completeness should be separated formally. The manuscript would become a more defensible conceptual contribution if its principal conclusion were moderated along the following lines:

> Operational physics does not independently justify treating exact instantaneous states as empirically accessible complete descriptions; consequently, the use of pointwise Markov or divisible models should be recognised as an idealisation whose adequacy must be assessed for each system.

The stronger conclusions–that instantaneous states are physically incomplete, CK indivisibility is generic, quantum behaviour is the default, and the measurement problem is dissolved–would require additional mathematical results or explicit dynamical models. Substantial revision is therefore recommended.

2 Major comments

The abstract states the conclusions more strongly than the arguments establish. The ε-δ definition does not guarantee that an instantaneous state lacks properties; it defines the limit independently of the value assigned to the original function at the limiting argument. The limit L is itself a well-defined mathematical object

and may belong to a state space carrying richer structure. The claim that quantum mechanics and general relativity guarantee that the limiting process physically self-destructs at the Planck scale should likewise be qualified. Minimum-length arguments are influential heuristic consequences of combining quantum and gravitational reasoning, but they are not established results internal to a completed theory of quantum gravity. The abstract's conclusion that indivisibility is consequently the generic expectation should therefore be presented as a proposed interpretation rather than as a consequence already demonstrated.

Section 2 assumes that a physical state at time t is necessarily the limit of interval-averaged states. States in classical mechanics, quantum mechanics, and field theory are commonly introduced directly as point-indexed mathematical objects, such as a trajectory $x(t)$, a density operator $\rho(t)$, or Cauchy data. Physical measurements may average these objects over nonzero intervals, but the measurement procedure should not be identified without further argument with the mathematical definition of the state.

The role of the punctured-neighbourhood condition $0 < |x - a| < \delta$ should also be reconsidered. Its purpose is to make the limit independent of a separately assigned value $f(a)$; it does not imply that the limit value L lacks structure or information. If the state is defined by the limit, then L, rather than the original value $f(a)$, is the relevant point-state. The example $\sin(x)/x$ shows only that a function and its continuous extension are distinct objects, not that an instantaneous physical state cannot be defined through such an extension.

The Landau-Lifshitz observation concerning the failure of a pointwise velocity along quantum trajectories does not establish the broader incompleteness of the quantum state. Nonexistence of a derivative of a trajectory and incompleteness of a state are logically distinct. Section 2 should therefore distinguish more carefully between pointwise kinematic quantities, mathematical state descriptions, and their operational accessibility.

Equation (1) and the associated discussion of CK divisibility would benefit from a more explicit mathematical setting. CK composition depends on the class of transition maps and generally involves summation or integration over an intermediate state. The manuscript should define the state space, the transition kernels, and the precise meaning of divisibility used here before inferring physical consequences from the factorisation condition. In particular, a separate argument is needed to show why limited operational access to an intermediate-time state prevents the corresponding transition maps from composing mathematically.

Sections 2 and 3 move from the limited claim that the construction of a point-state does not guarantee completeness to the stronger assertion that established physics contradicts completeness. The absence of a guarantee is not, by itself, evidence that the relevant property is false. This transition should be reformulated so that the evidential and deductive parts of the argument remain distinct.

The Mead-Garay discussion in Section 3 should also present the minimum-length conclusion with greater qualification. The argument provides a physically important estimate obtained by extrapolating quantum mechanics and classical general relativity into a regime in which neither is independently expected to remain complete. The discussion should distinguish minimum measurable resolution, minimum physically meaningful interval, spacetime discreteness, and failure of point-indexed mathematical states, since these propositions are not equivalent.

The treatment of the 't Hooft and Salecker-Wigner arguments in Section 3 requires some further qualification. Black-hole-formation arguments in trans-Planckian scattering do not by themselves establish a universal prohibition on all pointwise temporal descriptions. The Salecker-Wigner clock argument and the gravitational bound are also not fully independent once the Planck-scale estimate is obtained by combining the quantum clock constraint with avoidance of gravitational collapse. The manuscript should distinguish the result established by Salecker and Wigner from the subsequent gravitational combination and state the additional assumptions relating clock size, signal propagation, localisation, and collapse.

Even if a minimum operational time resolution is accepted, it follows only that the limit $\delta t \to 0$ cannot be physically implemented. Mathematical states and evolution maps routinely involve ideal quantities that cannot be measured with arbitrary precision. A further argument is required to show that this idealisation invalidates CK composition.

The treatment of renormalisation in Section 4 is too broad in several places. Quantum fields are operator-valued distributions, and products of fields at coincident points may be singular, but it is not generally correct to state that all physics evaluated at a spacetime point is infinite. Similarly, the statement that the point-valued limit is never taken should be qualified: regulators may be removed and continuum limits may exist once renormalised quantities have been defined. Ultraviolet divergences in intermediate expressions do not

by themselves show that spacetime points lack physical information.

The Wilsonian interpretation establishes scale dependence and the effective character of descriptions, but it does not by itself demonstrate that a bare theory at a spacetime point is always physically meaningless. On a fixed lattice, quantities are finite; renormalisation becomes necessary in relating descriptions across scales and, in particular, in taking a continuum limit. The lattice discussion should be revised accordingly.

The algebraic-QFT statement also needs its assumptions and implications stated more narrowly. Under suitable conditions, the intersection of observable algebras assigned to shrinking open regions may be trivial, but this concerns strictly point-local bounded observables. It does not establish that a state on a Cauchy surface is incomplete or that temporal propagators fail to compose.

Section 5 identifies a potentially useful criticism of formulations that seek a unique instantaneous moment of collapse, but it does not dissolve the measurement problem more generally. Decoherence describes the suppression of observable interference and the emergence of stable pointer bases; it does not by itself imply that, at the end of the measurement interaction, the apparatus has selected one definite outcome. That is part of the definite-outcome problem itself.

Rejecting an instantaneous moment of collapse also does not remove the broader tension between unitary evolution and definite outcomes, and collapse dynamics may be formulated continuously. The manuscript acknowledges that the Born-rule, preferred-basis, and definite-outcome problems remain untreated; given those exclusions, the section title and claims should be narrowed to the timing formulation of collapse rather than the measurement problem as a whole.

The discussion of the Barandes framework should also state precisely the class of systems and assumptions covered by the cited correspondence results. Claims of empirical equivalence and the derivation of quantum structures should be limited to the scope actually established there.

The condition $\delta t_{\mathrm{obs}} \gg \tau_{\mathrm{coh}}$ in Section 6 is introduced without a derivation or sufficiently precise definitions. No model is supplied showing that temporal coarse-graining restores CK divisibility under this condition. Coarse-graining may produce approximately Markovian dynamics in some settings, but it can also preserve or generate effective memory. The manuscript should either derive this assertion within an explicit model or present it as a qualitative possibility.

The proposed falsifiable prediction is also not yet distinctive. CK violations in financial, biological, or turbulent data may reflect classical non-Markovian memory, hidden variables, nonstationarity, or an insufficiently specified state space. Such violations do not by themselves indicate quantum behaviour or the incompleteness of instantaneous states. A quantitative criterion distinguishing the proposed mechanism from ordinary classical non-Markovianity would be needed to make the prediction discriminating.

Section 7 repeats the claim that the preceding arguments deny completeness, whereas Section 2 establishes only that completeness is not guaranteed by the limit construction. This shift from absence of proof to disproof should be corrected throughout the abstract, Introduction, and Discussion so that the strength of the conclusion remains consistent.

The references are generally relevant, but several claims should be matched more closely to the precise scope and conditions of the cited results. In particular, the manuscript should verify the exact Salecker-Wigner bound, the assumptions behind the algebraic-QFT point-algebra statement, and the interpretation of the financial CK test. The discussion would also benefit from engagement with work distinguishing operational from ontological spacetime limits, the distributional character of quantum fields, process-tensor approaches, and classical versus quantum non-Markovianity. This would help separate finite resolution, memory, and quantum indivisibility more clearly.